Chen Cheng

✉: chcheng@illinois.edu
Homepage: moriartycc.github.io

Employment

University of Illinois Urbana-Champaign, IL, USA

- **Assistant Professor**, Department of Statistics. 8/2026 – Current
- **Affiliated Positions**
  - **Core Member**, NSF-Simons AI Institute for the Sky (SkAI Institute). 8/2026 – Current

University of Chicago, IL, USA

- **Postdoctoral Scholar**, Department of Statistics. 7/2025 – 7/2026
  Advised by: Rina Foygel Barber.

Education

Stanford University, CA, USA

- **Ph.D. in Statistics**, Department of Statistics.
  Jointly advised by: John Duchi and Andrea Montanari 9/2019 – 6/2025
- GPA: 4.20/4.30.

Peking University, Beijing, China

- **B.S. in Computational Mathematics**, School of Mathematical Sciences. 9/2015 – 7/2019
- GPA: 3.90/4.00, Rank: 1/18. (with distinction)

Research Interests

Machine learning theory, Random matrix theory, High dimensional statistics, Reinforcement learning.

Publications

P16 **C. Cheng**, R. Liang, R. F. Barber. "Rolling Conformal Prediction in Sequential Model Training", 2026. arXiv.

P15 Y. Zhang, **C. Cheng**, S. Feng, Z. Zhou. "Distributionally Robust Bidding in First-Price Auctions", 2026. Accepted to the 65th IEEE Conference on Decision and Control (CDC), 2026. Full.

P14 **C. Cheng**, R. F. Barber. "Is Memorization Helpful or Harmful? Prior Information Sets the Threshold", 2026. Proceedings of the Thirty-Ninth Conference on Learning Theory (COLT), PMLR 336:1399–1433, 2026. PMLR. arXiv. Presented in [T19, T20, T21].

P13 **C. Cheng**, R. F. Barber. "Concentration Inequalities for Exchangeable Tensors and Matrix-valued Data", 2026. arXiv.

P12 **C. Cheng**, J. Duchi. "Some Robustness Properties of Label Cleaning", 2025. Transactions on Machine Learning Research (TMLR), 2026. arXiv.

P11 M. Celentano, **C. Cheng**, A. Pananjady, KA. Verchand. "State evolution beyond first-order methods I: Rigorous predictions and finite-sample guarantees", 2025. arXiv.

P10 **C. Cheng**, J. Duchi, D. Levy. "Geometry, Computation, and Optimality in Stochastic Optimization", 2024. Accepted to Mathematics of Operations Research, 2025. arXiv.

P9 F. Areces, **C. Cheng**, J. Duchi, R. Kuditipudi. "Two Fundamental Limits for Uncertainty Quantification in Predictive Inference", 2024. Proceedings of the Thirty-Seventh Conference on Learning Theory (COLT), PMLR 247:186–218, 2024. PMLR.

P8 **C. Cheng**, G. Cheng, J. Duchi. "Collaboratively Learning Linear Models with Structured Missing Data", 2023. Conference on Neural Information Processing Systems (NeurIPS), 2023. arXiv. Presented in [T7].

P7 **C. Cheng**, A. Montanari. "Dimension Free Ridge Regression", 2023. Annals of Statistics, vol. 52, no. 6, pp. 2879-2912, 2024. arXiv. Presented in [T6, T9, T10, T11, T12, T12, T13, T14, T15, T16].

P6 **C. Cheng**, H. Asi, J. Duchi. "How Many Labelers Do You Have? A Closer Look at Gold-Standard Labels", 2022. Journal of the American Statistical Association, published online October 9, 2026. JASA. arXiv. Presented in [T3, T6, T9, T10, T11, T12, T13, T14, T15, T16].

P5 **C. Cheng**, J. Duchi, R. Kuditipudi. "Memorize to Generalize: on the Necessity of Interpolation in High Dimensional Linear Regression", 2022. Conference on Learning Theory (COLT), PMLR 178:5528–5560, 2022. arXiv. Presented in [T4].

P4 M. Celentano, **C. Cheng**, A. Montanari. "The High-dimensional Asymptotics of First Order Methods with Random Data", 2021. Accepted to Annals of Applied Probability, 2026. arXiv. Presented in [T8].

P3 S. Cen, **C. Cheng**, Y. Chen, Y. Wei, Y. Chi, "Fast Global Convergence of Natural Policy Gradient Methods with Entropy Regularization", 2020. Operations Research 70(4):2563–2578, 2022. arXiv.

P2 **C. Cheng**, Y. Wei, Y. Chen, "Tackling Small Eigen-gaps: Fine-Grained Eigenvector Estimation and Inference under Heteroscedastic Noise", 2020. IEEE Transactions on Information Theory, 2021. arXiv.

P1 Y. Chen, **C. Cheng**, J. Fan, "Asymmetry Helps: Eigenvalue and Eigenvector Analyses of Asymmetrically Perturbed Low-Rank Matrices", 2019. Annals of Statistics, vol. 49, no. 1, pp. 435-458, 2021. arXiv. Presented in [T2].

Dissertation

"High dimensionality in modern machine learning: a random matrix theory perspective". Stanford University ProQuest Dissertations & Theses. 2025.

Academic Honors & Awards

Fellowship & Scholarship

- William R. Hewlett Stanford Graduate Fellowship. 09/2019
- Yizheng Special Scholarship and Merit Student. 09/2018
- Leo-KoGuan Scholarship and Merit Student. 09/2017
- National Scholarship and Merit Student Pacesetter (Highest Honor) . 09/2016

Awards

- George E. Nicholson Student Paper Competition. Finalist. 10/2021
- Excellent Graduate of Beijing. 06/2019
- Excellent Graduate of Elite Training Program of Applied Mathematics. 06/2019
- S.T. Yau College Student Mathematics Contests – Group (Probability). Silver Medalist. 08/2017
- Elite Training Program of Applied Mathematics & Pure Mathematics. 2016-2019
- Chinese Mathematical Olympiad (CMO). Gold Medalist. 2014
- National Olympiad in Informatics (NOI), China. Silver Medalist. 2012

Talks and Presentations

T21	[P14] IMS New Researcher Conference, UMass Amherst.	07/2026
T20	[P14] Conference on Learning Theory (COLT), San Diego.	07/2026
T19	[P14] INFORMS Optimization Society Conference, Atlanta.	03/2026
T18	Workshop in Frontiers in Learning Under Data Heterogeneity, University of Florida.	01/2026
T17	NSF-Simons NITMB MathBio Convergence Conference, Chicago.	08/2025
T16	[P6, P7], Statistics Seminar, University of Illinois Urbana-Champaign.	01/2025
T15	[P6, P7], Statistics Seminar, Rutgers University New Brunswick.	01/2025
T14	[P6, P7], Statistics Seminar, New York University Stern.	01/2025
T13	[P6, P7], Statistics Seminar, University of Cambridge DPMMS.	01/2025
T12	[P6, P7], Symposium on Statistics and Biostatistics, Peking University.	12/2024
T11	[P6, P7], Statistics Seminar, Tsinghua University YMSC.	12/2024
T10	[P6, P7], Statistics Seminar, Chinese University of Hong Kong.	12/2024
T9	[P6, P7], Statistics Seminar, National University of Singapore.	12/2024
T8	[P4], Simons Seminar, University of California Berkeley.	11/2024
T7	[P8], Conference on Neural Information Processing Systems (NeurIPS), New Orleans.	12/2023
T6	Stanford Stats Department Industrial Affiliated Conference, Stanford University.	11/2023
T5	[P6], Stanford Stats Department Industrial Affiliated Conference, Stanford University.	11/2022
T4	[P5], Conference on Learning Theory (COLT), London.	07/2022
T3	[P6], Stanford Data Analytics for What's Next (DAWN) workshop, Aptos.	05/2022
T2	[P1], Seminar for Modeling & Simulation, New York University Courant.	10/2018
T1	Seminar for Elite Ph.D Training Program of Applied Mathematics, Peking University.	11/2017

Teaching Experiences

As Instructor (UIUC)

- STAT 410. Statistics and Probability II. Fall 2026

As Teaching Assistant (Stanford)

- STATS 141. Biostatistics. Autumn 2019
- STATS 214/CS 229M. Machine Learning Theory. (Head TA) Winter 2021
- STATS 205. Introduction to Nonparametric Statistics. Spring 2021
- STATS 369. Methods from Statistical Physics. Autumn 2021
- MATH 230A/STATS 310A. Theory of Probability. Autumn 2022
- STATS 208. Bootstrap, Cross-Validation, and Sample Re-use. Winter 2023
- STATS 311/EE 377. Information Theory and Statistics. Autumn 2023
- STATS 315A. Modern Statistical Learning. Winter 2024
- STATS 118. Theory of Probability II. Summer 2024

Visiting and Working Experiences

LinkedIn Corporation, WA, USA

– **Research Intern**.
Supervised by: Ryan Rogers and Saikrishna Badrinarayanan 6/2023 – 9/2023

New York University, Courant Institute of Mathematical Sciences, NY, USA

– **Research Assistant**.
Supervised by: Miranda Holmes-Cerfon 9/2018 – 12/2018

Princeton University, NJ, USA

– **Research Assistant**.
Supervised by: Yuxin Chen 7/2018 – 9/2018

Professional Services

Reviewing

Journal

The Annals of Statistics (AoS), Journal of the American Statistical Association (JASA), Journal of the Royal Statistical Society, Series B: Statistical Methodology (JRSSB), Journal of Machine Learning Research (JMLR), IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE TPAMI), IEEE Transactions on Information Theory (IEEE TIT), Bernoulli, Foundations of Computational Mathematics (FoCM), Transactions on Machine Learning Research (TMLR), SIAM Journal on Mathematics of Data Science (SIMODS), Information and Inference: A Journal of the IMA (Inf. Inference), Mathematical Statistics and Learning (MSL), Electronic Communications in Probability (ECP)

Conference

Conference on Neural Information Processing Systems (NeurIPS), International Conference on Machine Learning (ICML), International Conference on Learning Representations (ICLR), ACM Symposium on Theory of Computing (STOC), Conference on Learning Theory (COLT), IEEE International Symposium on Information Theory (ISIT), International Conference on Algorithmic Learning Theory (ALT)